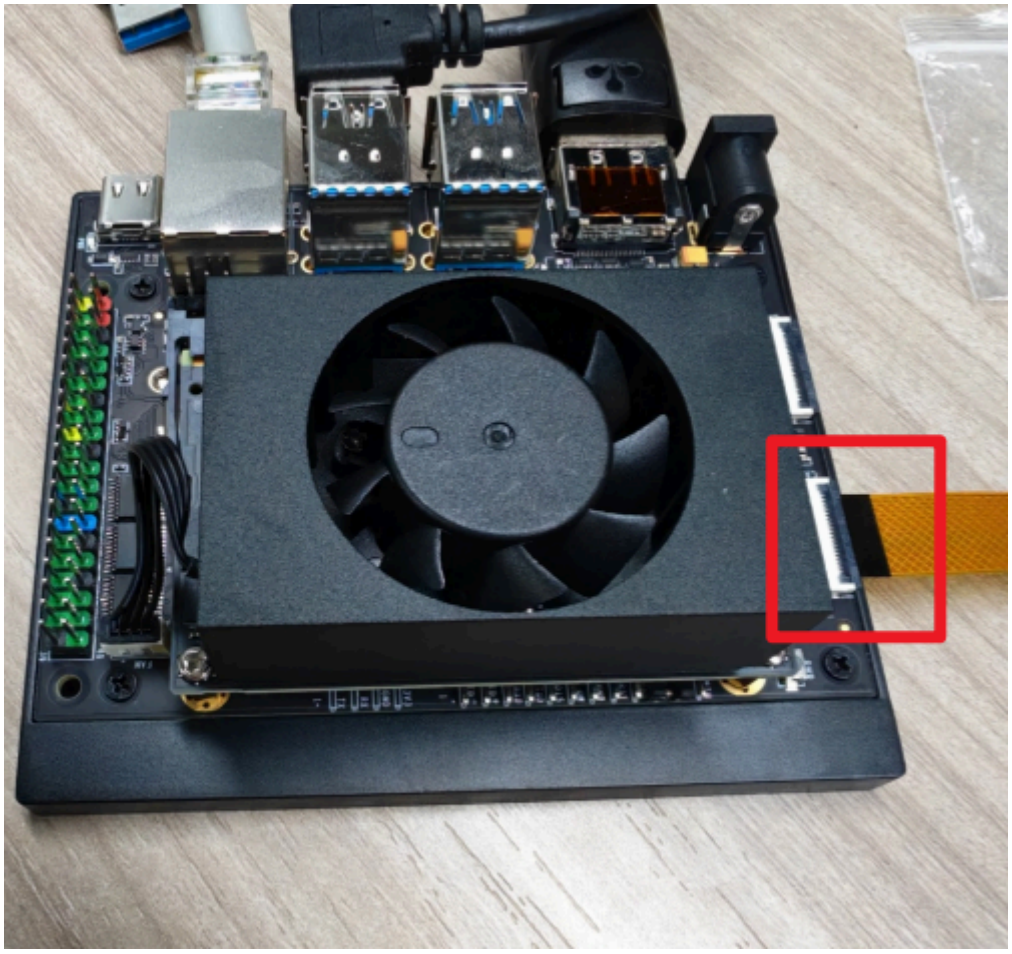
CSI Camera Preview

CSI Camera Preview
1. Enable the CSI Camera
    1.1. Configure the Pins
2. Check the Video Device
3. Preview the Camera Image
    3.1. Specify the Camera
    3.2. Specify the Preview Resolution

1. Enable the CSI Camera

Taking the IMX219 camera as an example, note that the golden fingers of the ribbon cable must face down during installation. Power off before installing or removing it; otherwise, the hardware could be damaged.

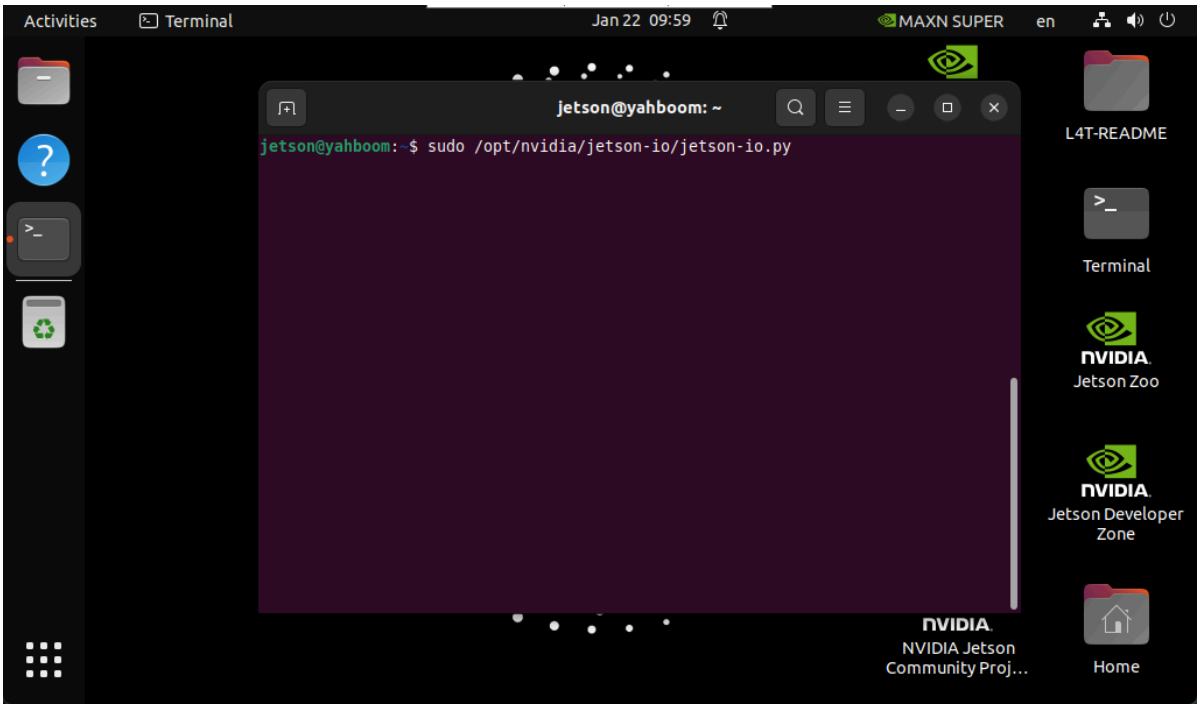


If the CSI camera is installed but there is no `/dev/video*` device, you can enable the CSI camera pins as follows

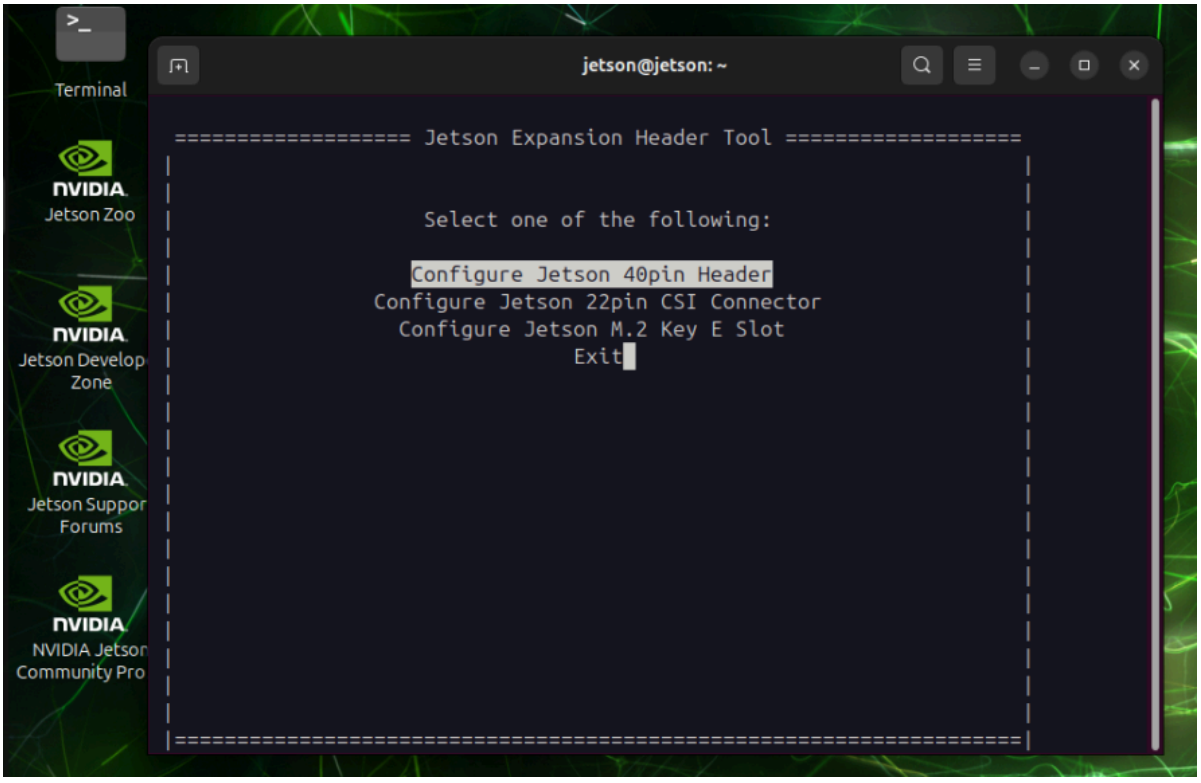
1.1. Configure the Pins

Enter the following command in the terminal:

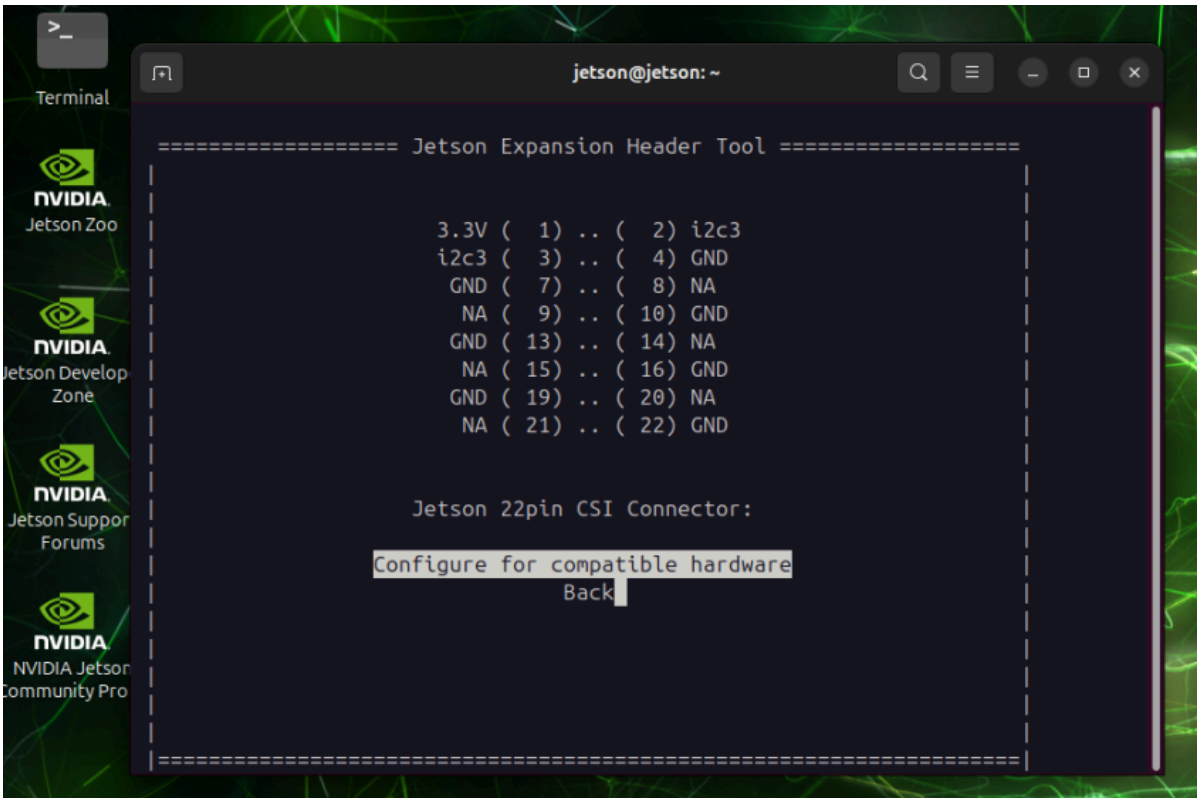
```
sudo /opt/nvidia/jetson-io/jetson-io.py
```



Press the arrow keys to select `Configure Jetson 22pin CSI Connector`, then press `Enter` to enter the configuration:



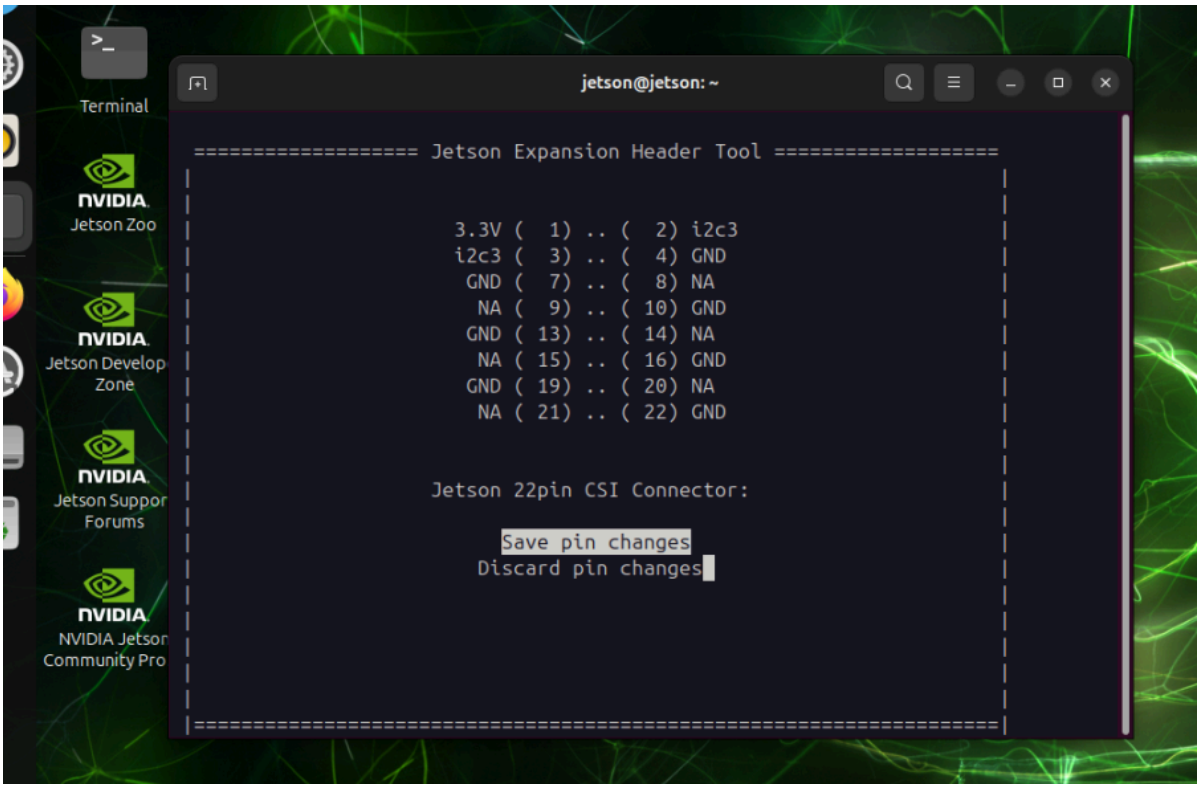
Select `Configure for compatible hardware` , then press `Enter` to enter the configuration:



Select `Camera IMX219 Dual` , then press `Enter` to enter the configuration:



Select `Save pin changes` , then press `Enter` to enter the configuration:



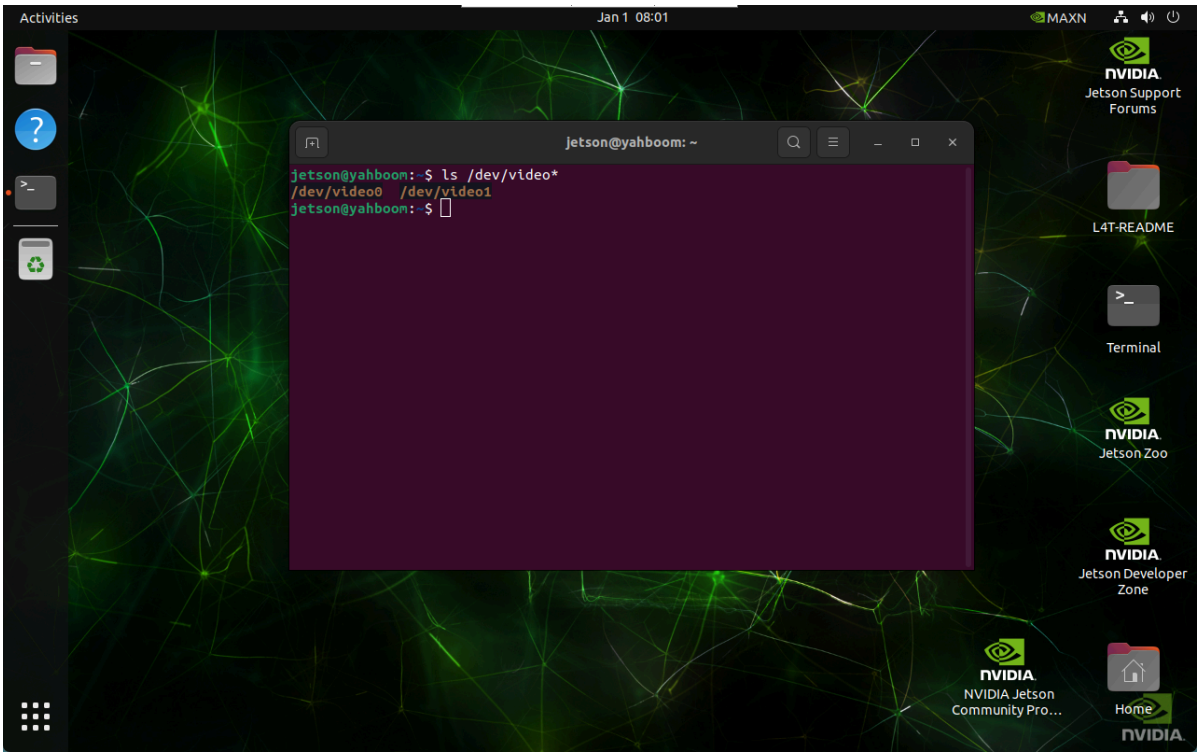
Select `Save and reboot to reconfigure pins` , then wait for the system to reboot:



2. Check the Video Device

```
ls /dev/video*
```

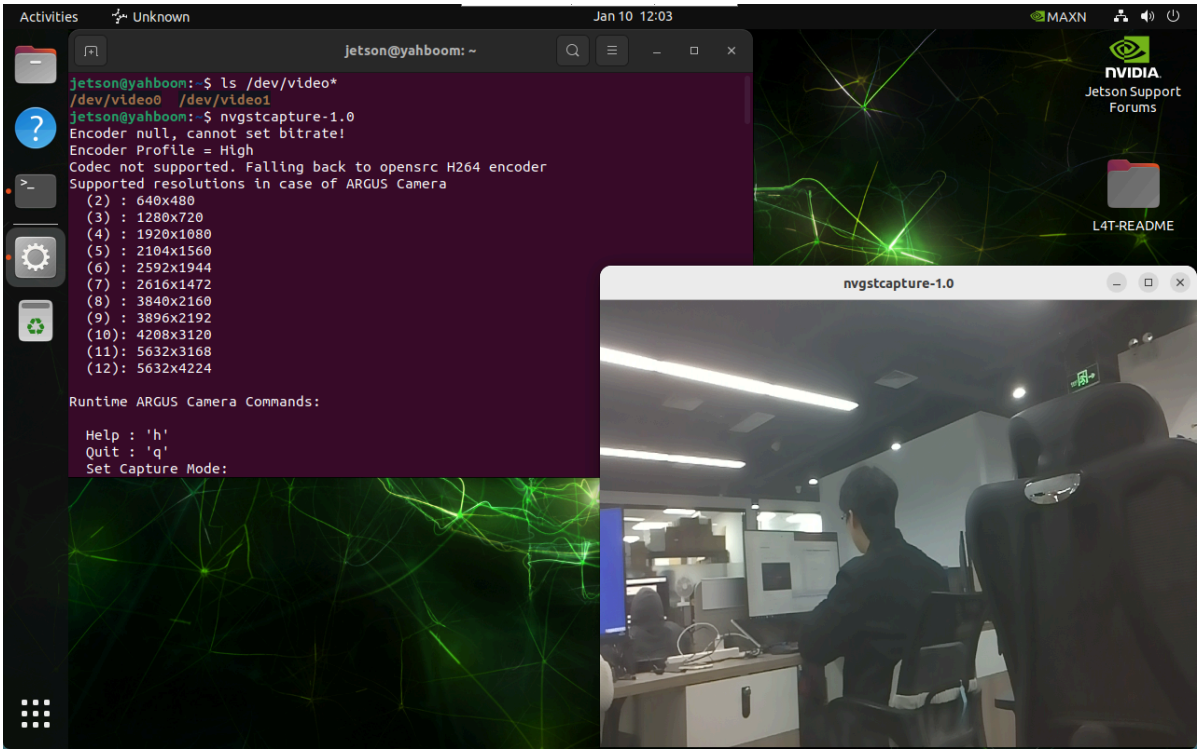
The result in the picture is the result of connecting two CSI cameras: generally, one CSI camera displays one `video` device



3. Preview the Camera Image

Enter the following command in the terminal, and the system automatically pops up the camera image window: the `/dev/video0` device is opened by default

```
nvgstcapture-1.0
```



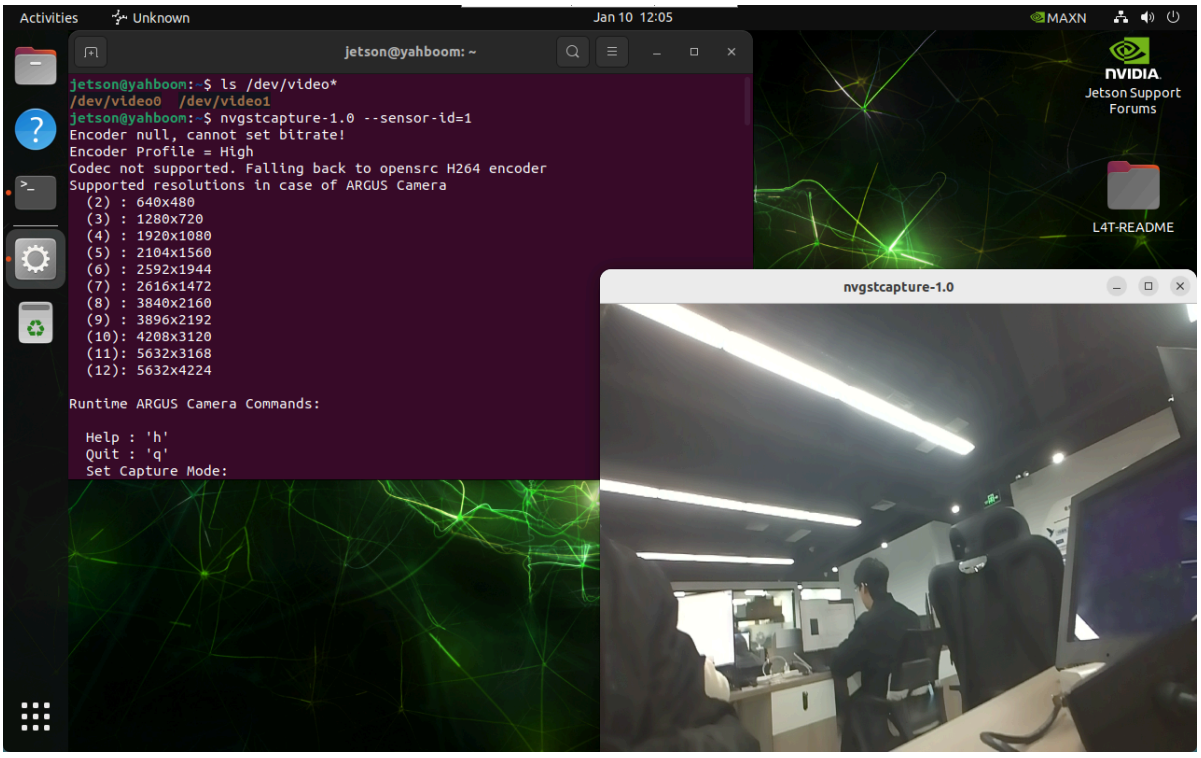
If the preview image is slightly reddish, you can add a parameter to adjust the white balance

```
nvgstcapture-1.0 --whitebalance=3
```

3.1. Specify the Camera

If there are multiple cameras, you can specify the camera ID:

```
nvgstcapture-1.0 --sensor-id=1
```



3.2. Specify the Preview Resolution

If there is only one CSI camera, `--sensor-id=1` can be changed to `--sensor-id=0`:

```
nvgstcapture-1.0 --sensor-id=1 --cus-prev-res=1280x720
```

